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SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		**DEPARTMENT OF COMPUTER SCIENCE ENGINEERING**
Program Name: B. Tech	**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal	
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CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week9 – Wednesday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
Assignment Number:17.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*Expected Time to complete*
1	**Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts**	Week9 - Wednesday

Lab Objectives:

- Learn how to clean raw datasets using AI-assisted Python scripting.
- Apply preprocessing techniques such as handling missing values, encoding categorical data, and normalization.
- Automate repetitive data-cleaning tasks with AI-generated code.
- Understand how preprocessing impacts model performance.

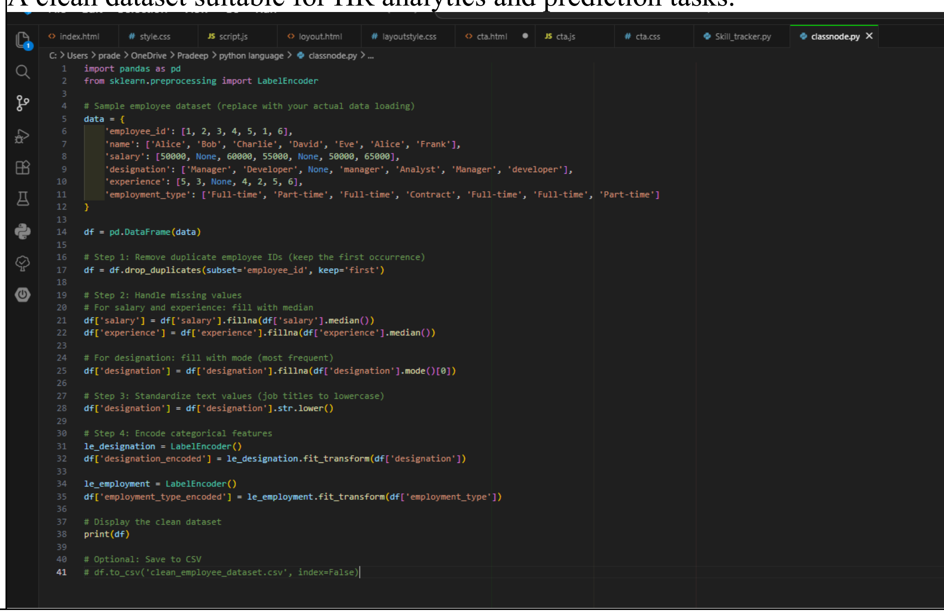
Task 1 Employee Dataset Cleaning

Task:
Use AI assistance to clean an employee information dataset.

Instructions:

- Handle missing values in salary, designation, and experience
- Remove duplicate employee IDs
- Standardize text values (job titles in lowercase)
- Encode categorical features (designation, employment_type)

Expected Output:
A clean dataset suitable for HR analytics and prediction tasks.



```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Sample employee dataset (replace with your actual data loading)
data = {
    'employee_id': [1, 2, 3, 4, 5, 1, 6],
    'name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Alice', 'Frank'],
    'salary': [50000, None, 60000, 55000, None, 50000, 65000],
    'designation': ['Manager', 'Developer', None, 'manager', 'Analyst', 'Manager', 'developer'],
    'experience': [5, 3, None, 4, 2, 5, 6],
    'employment_type': ['Full-time', 'Part-time', 'Full-time', 'Contract', 'Full-time', 'Full-time', 'Part-time']
}

df = pd.DataFrame(data)

# Step 1: Remove duplicate employee IDs (keep the first occurrence)
df = df.drop_duplicates(subset='employee_id', keep='first')

# Step 2: Handle missing values
# For salary and experience: fill with median
df['salary'] = df['salary'].fillna(df['salary'].median())
df['experience'] = df['experience'].fillna(df['experience'].median())

# For designation: fill with mode (most frequent)
df['designation'] = df['designation'].fillna(df['designation'].mode()[0])

# Step 3: Standardize text values (job titles to lowercase)
df['designation'] = df['designation'].str.lower()

# Step 4: Encode categorical features
le_designation = LabelEncoder()
df['designation_encoded'] = le_designation.fit_transform(df['designation'])

le_employment = LabelEncoder()
df['employment_type_encoded'] = le_employment.fit_transform(df['employment_type'])

# Display the clean dataset
print(df)

# Optional: Save to CSV
# df.to_csv('clean_employee_dataset.csv', index=False)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

```
Active code page: 65001

C:\Users\prade>C:/Users/prade/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/prade/OneDrive/Pradeep/python language/classnode.py"
   employee_id     name   salary designation  experience employment_type  designation_encoded  employment_type_encoded
0            1    Alice  50000.0     manager         5.0       Full-time                    2                        1
1            2      Bob  57500.0   developer         3.0       Part-time                    1                        2
Active code page: 65001

C:\Users\prade>C:/Users/prade/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/prade/OneDrive/Pradeep/python language/classnode.py"
   employee_id     name   salary designation  experience employment_type  designation_encoded  employment_type_encoded
0            1    Alice  50000.0     manager         5.0       Full-time                    2                        1
1            2      Bob  57500.0   developer         3.0       Part-time                    1                        2
   employee_id     name   salary designation  experience employment_type  designation_encoded  employment_type_encoded
0            1    Alice  50000.0     manager         5.0       Full-time                    2                        1
1            2      Bob  57500.0   developer         3.0       Part-time                    1                        2
0            1    Alice  50000.0     manager         5.0       Full-time                    2                        1
1            2      Bob  57500.0   developer         3.0       Part-time                    1                        2
1            2      Bob  57500.0   developer         3.0       Part-time                    1                        2
2            3  Charlie  60000.0     analyst         4.0       Full-time                    0                        1
2            3  Charlie  60000.0     analyst         4.0       Full-time                    0                        1
3            4    David  55000.0     manager         4.0        Contract                    2                        0
4            5      Eve  57500.0     analyst         2.0       Full-time                    0                        1
6            6    Frank  65000.0   developer         6.0       Part-time                    1                        2

C:\Users\prade>
```

Task 2 – Student Performance Data Preprocessing

Task:

Preprocess student performance data using AI assistance.

Instructions:

- Convert exam_date column into datetime format
- Extract new features (month, semester, day)
- Normalize total_marks
- Detect and handle outliers in marks using IQR or Z-score

Expected Output:

A transformed dataset with time-based academic features and normalized marks.

index.html # style.css JS script.js layout.html # layoutstyle.css cta.html JS cta.js # cta.css Skill_tracker.py

C: > Users > prade > OneDrive > Pradeep > python language > classnode.py > ...

```python

import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler

# Sample Data
np.random.seed(42)
dates = pd.date_range('2023-01-01', periods=100, freq='D')
df = pd.DataFrame({
    'student_id': range(1, 101),
    'exam_date': np.random.choice(dates, 100),
    'total_marks': np.random.normal(75, 15, 100),
})
df.loc[5, 'total_marks'] = 150  # Outlier

# 1. Convert to datetime
df['exam_date'] = pd.to_datetime(df['exam_date'])

# 2. Extract time features
df['month'] = df['exam_date'].dt.month
df['day'] = df['exam_date'].dt.day
df['semester'] = df['exam_date'].dt.month.apply(lambda x: 1 if x <= 6 else 2)

# 3. Outlier Detection - IQR Method
Q1, Q3 = df['total_marks'].quantile([0.25, 0.75])
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df['outlier'] = (df['total_marks'] < lower_bound) | (df['total_marks'] > upper_bound)

# 4. Outlier Detection - Z-Score Method
z_scores = np.abs((df['total_marks'] - df['total_marks'].mean()) / df['total_marks'].std())
df['z_outlier'] = z_scores > 3

# 5. Handle Outliers (Remove)
df = df[~df['outlier']].reset_index(drop=True)

# 6. Normalize Marks (Min-Max)
scaler = MinMaxScaler()
df['total_marks_normalized'] = scaler.fit_transform(df[['total_marks']])

# Display Results
print("Preprocessed Data:")
print(df[['student_id', 'exam_date', 'month', 'semester', 'total_marks', 'total_marks_normalized']].head(10))
print(f"\nOriginal shape: (100,), Processed shape: {df.shape}")
print(f"\nNormalized marks range: [{df['total_marks_normalized'].min():.4f}, {df['total_marks_normalized'].max():.4f}]")

# Save
df.to_csv(r'c:\Users\prade\OneDrive\Pradeep\python language\processed_student_data.csv', index=False)
print("\n✓ Processed data saved to processed_student_data.csv")
```

```
C:\Users\prade>C:/Users/prade/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/prade/OneDrive/Pradeep/python language/classnode.py"
Preprocessed Data:
   student_id  exam_date  month  semester  total_marks  total_marks_normalized
0           1 2023-02-21      2         1    54.799829                0.181188
1           2 2023-04-03      4         1    61.791131                0.306181
2           3 2023-01-15      1         1    58.041715                0.239148
3           4 2023-03-13      3         1    77.016433                0.578383
4           5 2023-03-02      3         1    83.731842                0.698443
5           7 2023-03-24      3         1    88.414985                0.782169
6           8 2023-03-28      3         1    86.324967                0.744804
7           9 2023-03-16      3         1    71.892512                0.486776
8          10 2023-03-16      3         1    65.647839                0.375132
9          11 2023-03-29      3         1    52.377701                0.137885

Original shape: (100,), Processed shape: (97, 9)
6           8 2023-03-28      3         1    86.324967                0.744804
7           9 2023-03-16      3         1    71.892512                0.486776
8          10 2023-03-16      3         1    65.647839                0.375132
9          11 2023-03-29      3         1    52.377701                0.137885

Original shape: (100,), Processed shape: (97, 9)
8          10 2023-03-16      3         1    65.647839                0.375132
9          11 2023-03-29      3         1    52.377701                0.137885

Original shape: (100,), Processed shape: (97, 9)

Original shape: (100,), Processed shape: (97, 9)
Original shape: (100,), Processed shape: (97, 9)


Normalized marks range: [0.0000, 1.0000]

✓ Processed data saved to processed_student_data.csv
```

Task 3 – Student Health Data Preparation

Task:

Clean and preprocess student health records using AI scripts.

Instructions:

- Handle missing values in height, weight, BMI
- Encode categorical features (medical_condition, fitness_status)
- Scale numerical features using standard scaling
- Split dataset into training and testing sets

Expected Output:

A preprocessed dataset ready for predictive health analysis.

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```python
# Generate Sample Health Data
np.random.seed(42)
df = pd.DataFrame({
    'student_id': range(1, 101),
    'age': np.random.randint(18, 25, 100),
    'height': np.random.normal(170, 10, 100),
    'weight': np.random.normal(70, 15, 100),
    'bmi': np.random.normal(24, 3, 100),
    'medical_condition': np.random.choice(['Healthy', 'Asthma', 'Diabetes', None], 100),
    'fitness_status': np.random.choice(['Excellent', 'Good', 'Fair', 'Poor', None], 100),
    'exercise_hours': np.random.uniform(0, 12, 100),
})

# Add missing values
missing_idx = np.random.choice(100, 10, replace=False)
df.loc[missing_idx[:3], 'height'] = np.nan
df.loc[missing_idx[3:6], 'weight'] = np.nan
df.loc[missing_idx[6:], 'bmi'] = np.nan

print("Before preprocessing:")
print(f"Missing values:\n{df.isnull().sum()}\n")

# 1. Handle Missing Values
df['height'].fillna(df['height'].mean(), inplace=True)
df['weight'].fillna(df['weight'].mean(), inplace=True)
df['bmi'].fillna(df['bmi'].mean(), inplace=True)
df['medical_condition'].fillna('None', inplace=True)
df['fitness_status'].fillna('Fair', inplace=True)

# 2. Encode Categorical Features
le_medical = LabelEncoder()
le_fitness = LabelEncoder()
df['medical_encoded'] = le_medical.fit_transform(df['medical_condition'])
df['fitness_encoded'] = le_fitness.fit_transform(df['fitness_status'])

# 3. Scale Numerical Features
numerical_cols = ['age', 'height', 'weight', 'bmi', 'exercise_hours']
scaler = StandardScaler()
df_scaled = df.copy()
df_scaled[numerical_cols] = scaler.fit_transform(df[numerical_cols])

# 4. Split Dataset
X = df_scaled[numerical_cols + ['medical_encoded', 'fitness_encoded']]
y = pd.cut(df['bmi'], bins=3, labels=['Normal', 'Overweight', 'Obese'])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("After preprocessing:")
print(f"Missing values: {df_scaled.isnull().sum().sum()}\nEncoded features: 2\nScaled features: 5")
print(f"\nTraining set: {len(X_train)}, Testing set: {len(X_test)}")
print(f"\n{X_train.head()}")

# Save
df_scaled.to_csv(r'c:\Users\prade\OneDrive\Pradeep\python language\processed_health_data.csv', index=False)
X_train.to_csv(r'c:\Users\prade\OneDrive\Pradeep\python language\health_train.csv', index=False)
X_test.to_csv(r'c:\Users\prade\OneDrive\Pradeep\python language\health_test.csv', index=False)

print("\n✓ Data saved: processed_health_data.csv, health_train.csv, health_test.csv")
```

```
C:\Users\prade>C:/Users/prade/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/prade/OneDrive/Pradeep/python language/classnode.py"
Before preprocessing:
Missing values:
student_id           0
age                  0
height               3
weight               3
bmi                  4
medical_condition   21
fitness_status      25
exercise_hours       0
dtype: int64
```

```
See the documentation for a more detailed explanation: https://pandas.pydata.org/pandas-docs/stable/user_guide/copy_on_write.html
  df['fitness_status'].fillna('Fair', inplace=True)
After preprocessing:
Missing values: 56
Encoded features: 2
Scaled features: 5

Training set: 80, Testing set: 20

         age    height    weight       bmi  exercise_hours  medical_encoded  fitness_encoded
55  1.465198  0.306841  0.023128  0.300740       -0.473295                1                2
88  0.414877 -1.596923 -0.885905 -0.042759        0.560665                3                1
26 -0.110284  0.412532 -0.387245 -0.502624        0.135869                3                4
42  1.465198 -0.817047 -0.364701  2.263329       -1.788820                1                0
69  1.465198  0.004662  1.032667  1.151545       -1.403043                2                2

✓ Data saved: processed_health_data.csv, health_train.csv, health_test.csv
After preprocessing:
Missing values: 56
Encoded features: 2
Scaled features: 5

Training set: 80, Testing set: 20

         age    height    weight       bmi  exercise_hours  medical_encoded  fitness_encoded
55  1.465198  0.306841  0.023128  0.300740       -0.473295                1                2
88  0.414877 -1.596923 -0.885905 -0.042759        0.560665                3                1
26 -0.110284  0.412532 -0.387245 -0.502624        0.135869                3                4
42  1.465198 -0.817047 -0.364701  2.263329       -1.788820                1                0
69  1.465198  0.004662  1.032667  1.151545       -1.403043                2                2

✓ Data saved: processed_health_data.csv, health_train.csv, health_test.csv
42  1.465198 -0.817047 -0.364701  2.263329       -1.788820                1                0
69  1.465198  0.004662  1.032667  1.151545       -1.403043                2                2

✓ Data saved: processed_health_data.csv, health_train.csv, health_test.csv
✓ Data saved: processed_health_data.csv, health_train.csv, health_test.csv
```

Task 4: Real-Time Application: Ride-Sharing Data Cleaning

Scenario:

A ride-sharing company has trip data with errors.

Requirements:

- Standardize pickup and drop locations
- Fill missing fare amounts with median values
- Remove duplicate ride entries
- Normalize driver ratings
- Generate a summary of data improvements

Expected Output:

A clean dataset ready for operational analysis.

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```python
# Sample Ride-Sharing Data
np.random.seed(42)
data = {
    'ride_id': range(1, 101),
    'pickup_location': np.random.choice(['  downtown  ', 'AIRPORT', 'Downtown', 'airport', None], 100),
    'drop_location': np.random.choice(['  midtown  ', 'STATION', 'station', 'Midtown', None], 100),
    'fare_amount': np.random.normal(25, 10, 100),
    'driver_rating': np.random.uniform(1, 5, 100),
}

df = pd.DataFrame(data)
# Add duplicates and missing values
df.loc[random_rows := np.random.choice(100, 5, replace=False), 'fare_amount'] = np.nan
df = pd.concat([df, df.iloc[:5]], ignore_index=True)

print("RIDE-SHARING DATA CLEANING PIPELINE")
print("="*60)
print(f"\nOriginal shape: {df.shape}")
print(f"Missing values:\n{df.isnull().sum()}\n")

# 1. Standardize Locations
df['pickup_location'] = df['pickup_location'].str.strip().str.lower().str.title()
df['drop_location'] = df['drop_location'].str.strip().str.lower().str.title()
df['pickup_location'].fillna('Unknown', inplace=True)
df['drop_location'].fillna('Unknown', inplace=True)
print("✓ Standardized pickup and drop locations")

# 2. Fill Missing Fares with Median
median_fare = df['fare_amount'].median()
df['fare_amount'].fillna(median_fare, inplace=True)
print(f"✓ Filled missing fares with median: ${median_fare:.2f}")

# 3. Remove Duplicates
initial_rows = len(df)
df = df.drop_duplicates(subset=['ride_id', 'pickup_location', 'drop_location'], keep='first')
df = df.reset_index(drop=True)
duplicates_removed = initial_rows - len(df)
print(f"✓ Removed duplicate entries: {duplicates_removed} rows")

# 4. Normalize Driver Ratings (0-1 scale)
scaler = MinMaxScaler()
df['driver_rating_normalized'] = scaler.fit_transform(df[['driver_rating']])
print(f"✓ Normalized driver ratings (0-1 scale)")

# 5. Generate Summary
print("\n" + "="*60)
print("DATA QUALITY IMPROVEMENT SUMMARY")
print("="*60)
print(f"Final dataset shape: {df.shape}")
print(f"Duplicate entries removed: {duplicates_removed}")
print(f"Missing fares filled: {(df['fare_amount'].isna().sum() == 0)}")
print(f"Location standardization: Complete")
print(f"Driver ratings normalized: Complete")
print(f"\nFare Statistics:")
print(f"  Min: ${df['fare_amount'].min():.2f}")
print(f"  Max: ${df['fare_amount'].max():.2f}")
print(f"  Mean: ${df['fare_amount'].mean():.2f}")
print(f"  Median: ${df['fare_amount'].median():.2f}")
```

```python
# Save cleaned data
df.to_csv(r'c:\Users\prade\OneDrive\Pradeep\python language\cleaned_rides.csv', index=False)
print(f"\n✓ Clean dataset saved to cleaned_rides.csv")
print(f"\nCleaned Data Sample:\n{df.head()}")
```

```
C:\Users\prade>C:/Users/prade/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/prade/OneDrive/Pradeep/python language/classnode.py"
RIDE-SHARING DATA CLEANING PIPELINE
============================================================

Original shape: (105, 5)
Missing values:
ride_id             0
pickup_location    22
drop_location      17
fare_amount         5
driver_rating       0
dtype: int64

============================================================
DATA QUALITY IMPROVEMENT SUMMARY
============================================================
  df['fare_amount'].fillna(median_fare, inplace=True)
✓ Filled missing fares with median: $25.44
✓ Removed duplicate entries: 5 rows
✓ Normalized driver ratings (0-1 scale)

============================================================
DATA QUALITY IMPROVEMENT SUMMARY
============================================================
Final dataset shape: (100, 6)
Duplicate entries removed: 5
Missing fares filled: False
Location standardization: Complete
Driver ratings normalized: Complete

Fare Statistics:
  Min: $1.82
  Max: $55.61
  Mean: $25.20
  Median: $25.28
  df['fare_amount'].fillna(median_fare, inplace=True)
✓ Filled missing fares with median: $25.44
✓ Removed duplicate entries: 5 rows
✓ Normalized driver ratings (0-1 scale)

============================================================
DATA QUALITY IMPROVEMENT SUMMARY
============================================================
Final dataset shape: (100, 6)
Duplicate entries removed: 5
Missing fares filled: False
Location standardization: Complete
Driver ratings normalized: Complete

Fare Statistics:
  Min: $1.82
  Max: $55.61
Final dataset shape: (100, 6)
Duplicate entries removed: 5
Missing fares filled: False
Location standardization: Complete
Driver ratings normalized: Complete

Fare Statistics:
  Min: $1.82
  Max: $55.61
  Mean: $25.20
  Median: $25.28

✓ Clean dataset saved to cleaned_rides.csv

Fare Statistics:
  Min: $1.82
  Max: $55.61
  Mean: $25.20
  Median: $25.28

✓ Clean dataset saved to cleaned_rides.csv
  Mean: $25.20
  Median: $25.28

✓ Clean dataset saved to cleaned_rides.csv


 Cleaned Data Sample:

 Cleaned Data Sample:
   ride_id pickup_location drop_location  fare_amount  driver_rating  driver_rating_normalized
   ride_id pickup_location drop_location  fare_amount  driver_rating  driver_rating_normalized
0        1         Airport       Midtown    24.545973       2.312611                  0.323680
0        1         Airport       Midtown    24.545973       2.312611                  0.323680
1        2             NaN       Midtown    27.169367       1.620166                  0.144575
2        3        Downtown       Midtown    30.124237       4.927364                  1.000000
1        2             NaN       Midtown    27.169367       1.620166                  0.144575
2        3        Downtown       Midtown    30.124237       4.927364                  1.000000
3        4             NaN       Station    30.434887       4.355734                  0.852145
4        5             NaN       Midtown    25.280995       4.441618                  0.874359
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and if required, screenshots.